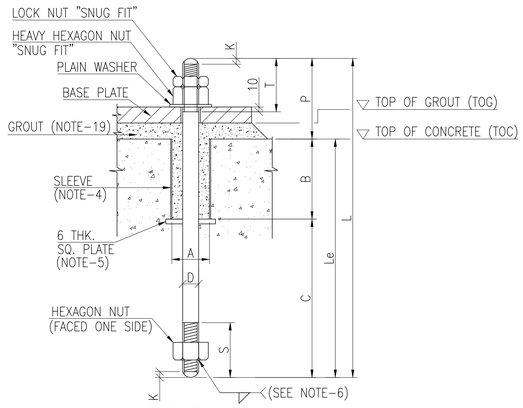
TITLE: STANDARD DRAWING FOR ANCHOR BOLT DETAILS

DWG NO: IPEG-6389-8310-39-002-0003

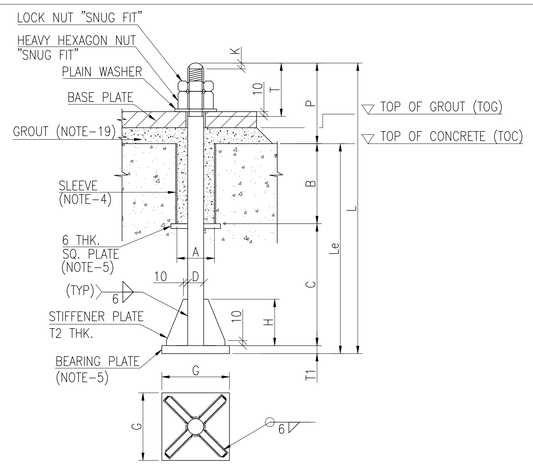
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TYPE A

TYPE A

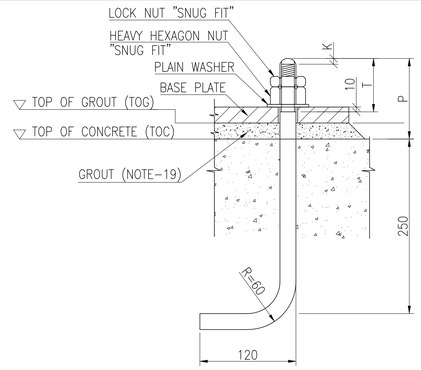
MARK	D (MM)	THREAD PITCH (MM)	NET AREA (NOTE-20) (MM^2)	MAX. PROJ.(P) (NOTE-16) (MM)	LENGTH OF BOLT L (MM)	T (MM)	K (MM)	C MIN. (NOTE-16) (MM)	Le MIN. (NOTE-16) (MM)	S (MM)	E MIN (NOTE-21) (MM)	F MIN (MM) (NOTE-21)			SLEEVE DETAILS (NOTE-4)		PERMISSIBLE ULTIMATE CAPACITY		PERMISSIBLE WORKING CAPACITY		APPROX. WT. OF A.BOLT ASSEMBLY (1 No.) (kg)
												STRUCTURAL STEEL WORK	HORIZONTAL DRUM / EXCHANGER	COLUMN, VERTICAL VESSEL / DRUM	A (MM)	B (MM)	TENSION kN	SHEAR kN	TENSION kN	SHEAR kN	
M 12	12	1.75	40	105	475	60	2.0	220	370	25	150	60	60	75	50	150	11	7	5	3	0.48
M 16	16	2.0	92	115	625	70	2.0	360	510	25	150	80	80	100	50	150	27	17	14	9	1.09
M 20	20	2.5	160	150	775	80	2.5	325	625	30	150	100	120	120	50	300	46	30	25	16	2.26
M 25	25	3.0	276	510	1350	95	3.0	440	840	35	150	125	125	150	80	400	80	51	43	28	5.54
M 30	30	3.5	423	525	1450	130	3.5	475	925	40	160	150	150	180	80	450	122	78	68	43	8.74
M 33.5	33.5	3.5	560	535	1550	140	3.5	565	1015	40	170	170	170	205	80	450	159	104	92	60	11.61
M 36	36	4.0	646	545	1650	150	4.0	655	1105	45	180	180	180	220	80	450	186	119	105	68	14.38



TYPE B

TYPE B

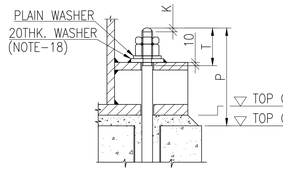
MARK	D (MM)	THREAD PITCH (MM)	NET AREA (NOTE-20) (MM^2)	MAX. PROJ.(P) (NOTE-16) (MM)	LENGTH OF BOLT L (MM)	T (MM)	K (MM)	C MIN. (NOTE-16) (MM)	Le MIN. (NOTE-16) (MM)	BOTTOM PLATE			T2 (MM)	E MIN (NOTE-21) (MM)	F MIN (MM) (NOTE-21)			SLEEVE DETAILS (NOTE-4)		PERMISSIBLE ULTIMATE CAPACITY		PERMISSIBLE WORKING CAPACITY		APPROX. WT. OF A.BOLT ASSEMBLY (1 No.) (kg)
										G (MM)	H (MM)	T1 (MM)			STRUCTURAL STEEL WORK	HORIZONTAL DRUM / EXCHANGER	COLUMN, VERTICAL VESSEL / DRUM	A (MM)	B (MM)	TENSION kN	SHEAR kN	TENSION kN	SHEAR kN	
M 40	40	4.0	838	560	1450	160	4.0	420	890	150	150	20	8	200	200	200	240	100	450	235	155	137	91	22.16
M 42	42	4.5	915	570	1525	170	4.0	485	955	150	150	20	8	210	210	210	255	100	450	261	169	150	97	24.65
M 45	45	4.5	1083	575	1575	175	5.0	530	1000	150	150	20	8	225	225	225	270	100	450	302	200	177	117	28.01
M 48	48	5.0	1231	670	1700	180	5.0	555	1030	200	175	25	8	240	240	240	290	100	450	347	227	201	132	38.90
M 50	50	5.0	1359	670	1775	185	5.0	630	1105	200	175	25	8	250	250	250	300	100	450	379	251	222	147	42.46
M 53	53	5.0	1562	690	1825	200	5.0	660	1135	200	175	25	8	265	265	265	320	125	450	428	289	256	172	47.01
M 56	56	5.5	1739	700	1925	210	6.0	750	1225	200	175	25	8	280	280	280	340	125	450	481	321	284	190	53.15
M 60	60	5.5	2047	765	2250	220	6.0	1007	1485	250	175	28	8	300	300	300	360	125	450	557	378	335	227	73.87
M 64	64	6.0	2334	780	2275	235	6.0	863	1495	250	200	32	8	320	320	320	385	150	600	638	431	382	258	84.68
M 71	71	6.0	2971	835	2350	260	7.0	883	1515	300	225	32	8	360	435	435	435	150	600	792	549	486	337	111.26
M 76	76	6.0	3474	850	2400	260	7.0	918	1550	300	225	32	8	380	450	450	460	150	600	913	642	568	400	124.38
M 80	80	6.0	3905	850	2500	260	7.0	1018	1650	300	225	32	8	400	480	480	480	150	600	1016	721	639	454	138.28



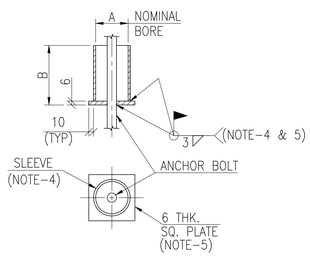
TYPE C

TYPE C

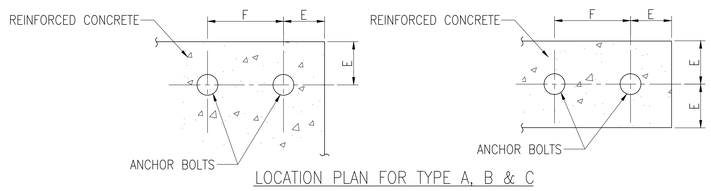
MARK	D (MM)	THREAD PITCH (MM)	NET AREA (NOTE-20) (MM^2)	MAX. PROJ.(P) (NOTE 16) (MM)	T (MM)	K (MM)	E MIN (NOTE-21) (MM)	F MIN (MM) (NOTE-21)			APPROX. WT. OF A.BOLT ASSEMBLY (1 No.) (kg)	REMARKS
								STRUCTURAL STEEL WORK	HORIZONTAL DRUM / EXCHANGER	COLUMN, VERTICAL VESSEL / DRUM		
M 20	20	2.50	160	105	80	2.5	150	100	100	120	1.40	THIS BOLT SHALL BE USED ONLY FOR SUPPORTING STAIR/LADDER AND MINOR PIPE SUPPORTS/OPERATING PLATFORMS/CROSS OVERS, ETC.



ANCHOR BOLT WITH CHAIR (NOTE-17)



SLEEVE DETAIL



LOCATION PLAN FOR TYPE A, B & C

LEGEND:

ANCHOR BOLTS SHALL BE MARKED ON DRAWING AS UNDER
No-C-MARK-PROJ
- PROJ — ACTUAL PROJECTION OF BOLTS ABOVE TOP OF CONCRETE
- MARK — MARK OF ANCHOR BOLT
- C — TYPE OF ANCHOR BOLT
- No — NUMBER OF ANCHOR BOLTS

e.g 2-C-M20-105

FOR NUMBER OF ANCHOR BOLTS & PROJECTION REFER TO DESIGN DRAWINGS.

NOTES:

1. ALL DIMENSIONS & SIZES INDICATED ARE IN MM.
2. FOR GENERAL NOTES OF CONCRETE WORKS AND STEEL WORKS REFER TO DRAWING NO IPEG-6389-8310-39-002-0001 AND IPEG-6389-8320-39-002-0001 RESPECTIVELY.
3. ANCHOR BOLT MATERIAL :
   (a) ANCHOR BOLTS SHALL BE TURNED FROM MS ROUNDS AND CONFIRM TO IS:1367 PART-3, MINIMUM GRADE 4.6 OR EQUIVALENT.
   (b) NUTS SHALL BE HEXAGON NUT (GRADE "C" AS PER IS:1363).
   (c) PLAIN WASHERS TO BE BLACK FORM "E" TO IS:6649.
   (d) SCREW THREADS TO BE I.S.O. METRIC COARSE SERIES, FREE FIT CLASS (7H/8G) TO ISO 724 (IS:4218-PART-3).
   (e) BOLT >64MM DIA. SHALL HAVE 6MM CONSTANT PITCH.
4. BOLT SLEEVE TO BE FABRICATED FROM CARBON STEEL SCHEDULE 40 UNGALVANIZED PIPE. SLEEVE SHALL BE SHIPPED LOOSE AND WELDED AT SITE WITH ANCHOR BOLT BASED ON ACTUAL PROJECTION AS SPECIFIED IN DESIGN DRAWING.
5. 6 THICK SQUARE PLATE SHALL BE ADEQUATELY TACK WELDED ON FOUR SIDES TO THE SLEEVE AND BOLT. ALL PLATE MATERIAL (INCLUDING BEARING PLATE) SHALL CONFIRM TO IS:2062.
6. NUT AT THE BOTTOM SHALL BE TACK WELDED AT TWO LOCATIONS.
7. ALL ANCHOR BOLTS SHALL BE SET ACCURATELY AND SECURELY HELD IN POSITION BY TEMPLATE, PRIOR TO AND DURING CONCRETING.
8. THE TOP ELEVATION OF BOLTS SHALL BE CHECKED BEFORE PLACEMENT OF CONCRETE. PROJECTION TOLERANCE SHALL BE +5MM AND -2MM FOR BOLTS UP TO M 24, +8MM AND -4MM FOR BOLTS M 30 AND ABOVE.
9. UPON REMOVAL OF TEMPLATE THE BOLTS SHALL BE CHECKED FOR PLAN LOCATION. ACCEPTABLE TOLERANCES ARE 3MM WITHOUT SLEEVE AND 6MM WITH SLEEVE IN ANY DIRECTION FROM TRUE LOCATION.
10. WHEN THE BOLTS HAVE SATISFIED THE ABOVE REQUIREMENTS, THEY SHALL BE THOROUGHLY CLEANED AND THE NECESSARY WASHER AND NUT(S) ADDED AS INDICATED ON THE DETAIL DRAWINGS. EXPOSED PARTS OF THE ANCHOR BOLTS SHALL BE PROTECTED BY APPLYING GREASE & COVERING WITH POLYETHLENE/JUTE BAG OR BY DENSOTAPE OR SIMILAR TILL ERECTION OF THE EQUIPMENT/STRUCTURE.
11. THE ALLOWABLE ULTIMATE CAPACITY FOR TENSION AND SHEAR OF ANCHOR BOLT ARE BASED ON Fu=400Mpa & Fy=240Mpa AS PER IS:1367.
12. ALLOWABLE ULTIMATE CAPACITY OF ANCHOR BOLT UNDER TENSION AND SHEAR ARE WORKED OUT BASED ON CLAUSE 10.3.5 & 10.3.3 OF IS:800-2007 RESPECTIVELY.
13. ALLOWABLE WORKING CAPACITY OF ANCHOR BOLT UNDER TENSION AND SHEAR ARE WORKED OUT BASED ON CLAUSE NO. 11.6.2.3 AND 11.6.2.1 OF IS:800-2007 RESPECTIVELY.
14. SHEAR AND TENSION SHALL BE CHECKED WITH FOLLOWING EXPRESSION AS PER CLAUSE 10.3.6 & 11.6.2.5 OF IS 800:2007.
$$\left(\frac{\text{ACTUAL SHEAR FORCE ON BOLT}}{\text{SHEAR CAPACITY}}\right)^2+\left(\frac{\text{ACTUAL TENSILE FORCE ON BOLT}}{\text{TENSILE CAPACITY}}\right)^2\leq 1.0$$
15. TENSION AND SHEAR CAPACITIES REPRESENTED IN THE TABLE ARE FOR LONG TERM LOADING. THESE CAPACITIES CAN BE INCREASED BY 25% FOR LOAD COMBINATIONS INVOLVING SHORT TERM LOADING (WIND OR SEISMIC LOADS)
16. ANCHOR BOLTS SHALL BE PROJECTED ABOVE TOC AS PER ACTUAL REQUIRED PROJECTION. HOWEVER MINIMUM EMBEDMENT LENGTH Le SHALL BE ENSURED USING C MIN.
17. ANCHOR BOLT WITH CHAIR MAY BE PROVIDED FOR ANCHOR BOLTS M25 AND ABOVE.
18. 20MM THICK CIRCULAR WASHER SHALL BE SHIPPED LOOSE AND WELDED AT SITE AFTER ANCHOR BOLTS ARE IN POSITION AND ALIGNMENT OF THE STRUCTURE.
19. FOR THICKNESS OF GROUT, INDIVIDUAL FOUNDATION DRAWINGS OF RESPECTIVE ITEM SHALL BE REFERRED.
    G1 : NON SHRINK CEMENTATIOUS : 30MM MIN. THK, 40Mpa
    G2 : EPOXY GROUT (AS PER VENDOR SPECIFICATION)
    G3 : CEMENTATIOUS GROUT : 25MM MIN. THK, 22Mpa
20. TENSION & SHEAR CAPACITIES OF ANCHOR BOLTS ARE EVALUATED BASED ON NET AREA OF BOLTS CONSIDERING 3MM CORROSION ALLOWANCE.
21. MINIMUM DISTANCE FROM THE CENTERLINE OF FOUNDATION/ANCHOR BOLT TO EDGE OF PEDESTAL SHOULD BE THE MAXIMUM OF THE FOLLOWING,AS PER CL. NO. 13 OF DOC. NO. PDRP-8310-SP-0003.
    (a) CLEAR DISTANCE FROM THE EDGE OF THE BASE PLATE UP TO SIZE 500MM TO OUTER EDGE OF THE PEDESTAL SHOULD BE MINIMUM 50MM.
    (b) CLEAR DISTANCE FROM THE EDGE OF THE BASE PLATE OF SIZE HIGHER THAN 500MM/BASE FRAME TO OUTER EDGE OF THE PEDESTAL SHOULD BE MINIMUM 75MM.
    (c) CLEAR DISTANCE FROM THE FACE OF POCKET TO THE OUTER EDGE OF THE PEDESTAL SHOULD BE 100MM.
    (d) CLEAR DISTANCE FROM THE EDGE OF THE SLEEVE OR ANCHOR PLATE TO THE EDGE OF OF PEDESTAL SHOULD BE 100MM.

Drawing/Document Category (Submission Purpose): Drawing/Document Review Status (BY OWNER/PMC)

- ☐ For Approval
- ☐ For Review
- ☑ For Information

APPROVAL CODE	STATUS	ACTION REQUIRED
1	Rejected	Revise and resubmit
2	Commented as noted	Work may proceed subject to acceptance and incorporation of comments.
3	Acceptable with minor comments	Incorporate comments, if any. Resubmit as certified final.Work may proceed.
4	Approved with No comments-FINAL	Work may proceed.
5	Information Only	Accepted for information only.Work may proceed.

Date: __/__/____ Name : _____

This approval does not absolve the CONTRACTOR from meeting their contractual obligations of delivering a complete safe and efficiently operating unit facilities meeting all requirements of the contract.

REV	PURPOSE	PREPARE	CHKD	APPROVAL	REVIEWED	APPROVED	DATE dd/mm/yy
I1	ISSUED FOR INFORMATION	ADC	YUK	CAM	-	RMK	06/04/19

PROJECT: ETHYLENE GLYCOL PROJECT PARADIP REFINERY, ODISHA-INDIA.

OWNER: INDIAN OIL CORPORATION LIMITED PARADIP REFINERY, ODISHA, INDIA

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PMC: Toyo Engineering India Private Limited, Mumbai, India
TEIPL's Job No. 6389 WORK ORDER No. 25574423

LSTK-2: L&T Hydrocarbon Engineering
LSTK-2 Job No. IOCL-O&U WORK ORDER No. 25856933

DETAILED ENGINEERING CENTER: L&T - CHIYODA LIMITED

TITLE: STANDARD DRAWING FOR ANCHOR BOLT DETAILS

UNIT NO-

SCALE NTS | DWG. NO. IPEG-6389-8310-39-002-0003 | REV. I1 | 1/1

CAD FILE REF: IPEG-6389-8310-39-002-0003_I1